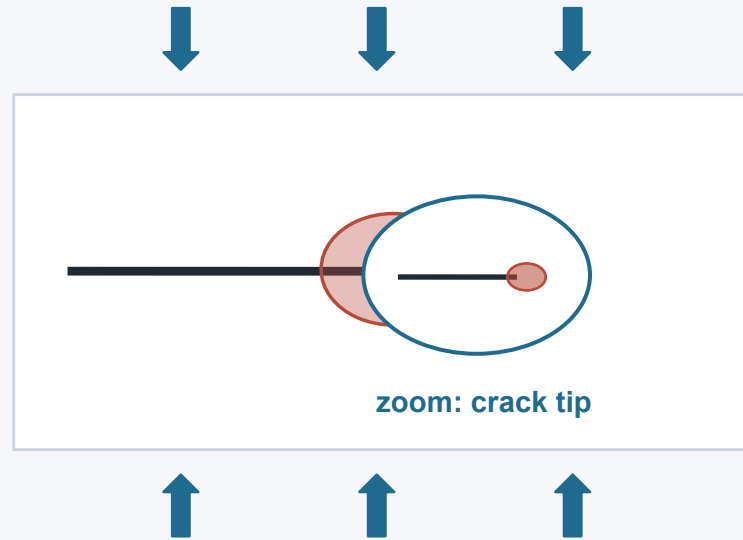


Start With a Crack

A component is loaded
A small crack exists
Failure starts at the crack tip

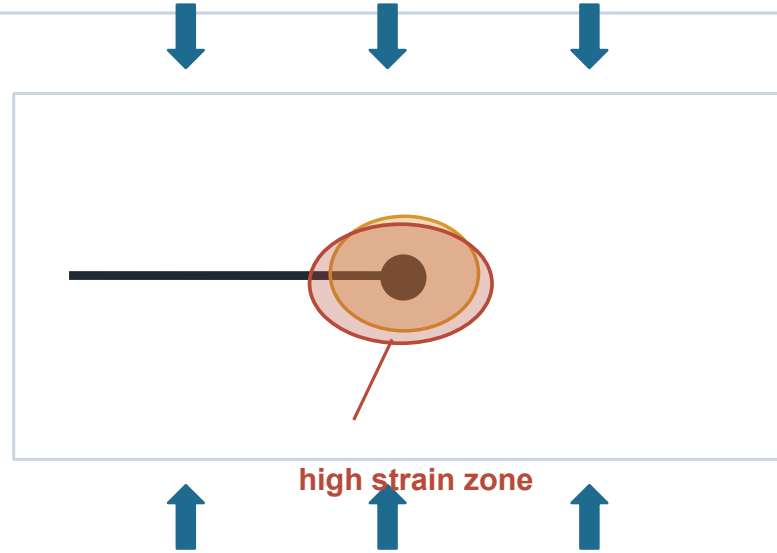


Why the Tip Is the Problem

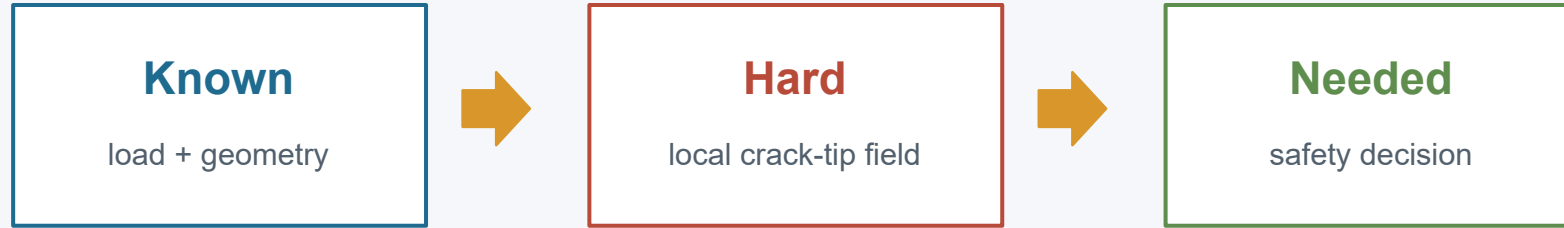
Far away: loading looks simple

Near the tip: deformation concentrates

Local yielding may begin first



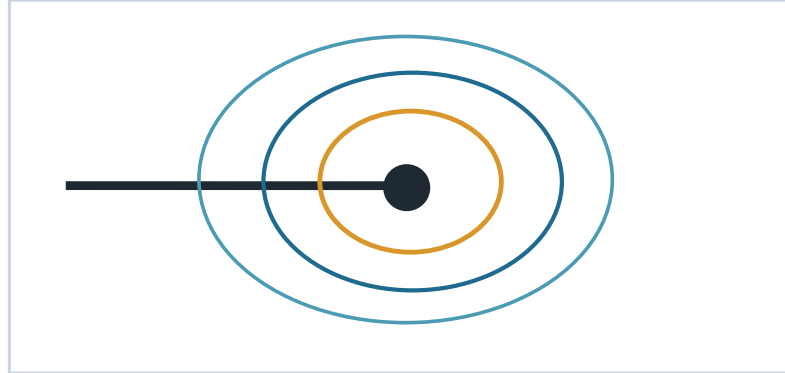
The Old Difficulty



Rice's question: can we avoid solving every local detail?

Rice's Simple Picture

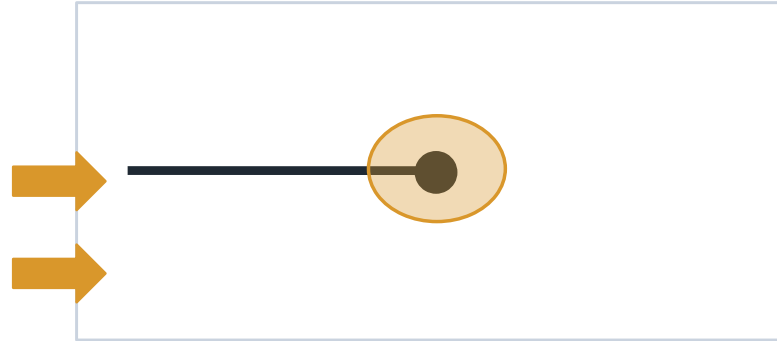
Draw a contour around the tip
Calculate J
Every valid path gives the same value



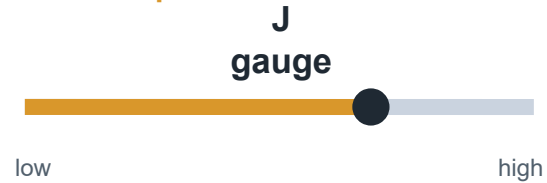
$$J1 = J2 = J3$$

What J Means in Plain Language

J is crack-driving energy
It is not one stress value at the tip
Higher J means stronger drive for crack growth



energy into crack tip



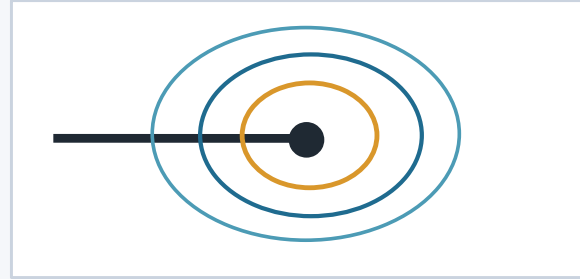
The Only Equation We Need

$$J = \text{contour integral} (W dy - T \cdot du/dx ds)$$

W: stored strain energy density

T: traction on the contour

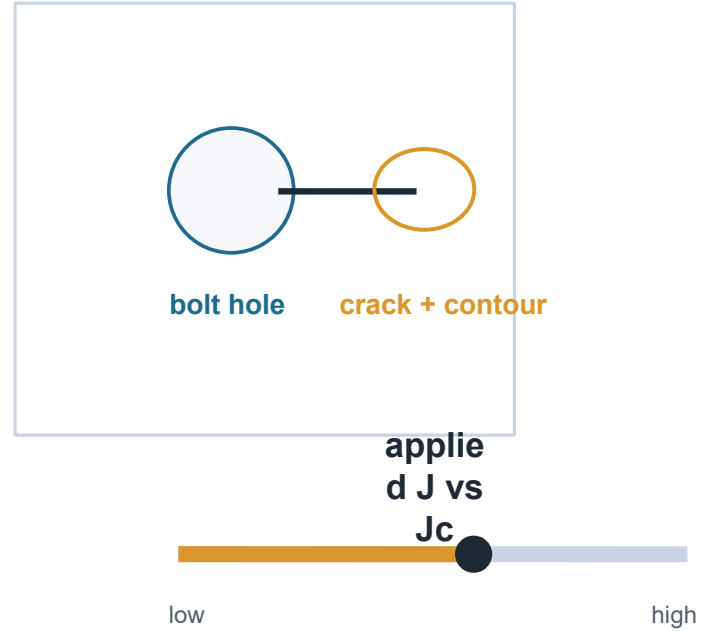
u: displacement field



contour around tip

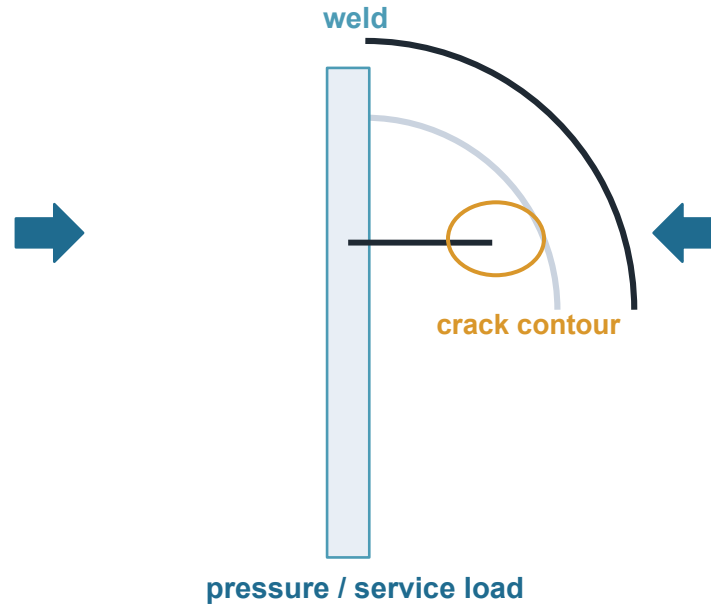
Case Study 1: Cracked Steel Bracket

Identify crack and load
Calculate J around the tip
Compare with material resistance J_c
Decide: monitor, repair, or redesign



Case Study 2: Welded Pipe Crack

Weld region raises local severity
Crack tip may yield locally
J handles nonlinear crack-driving force



What Rice Changed

Griffith / Irwin



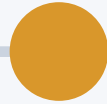
linear energy ideas

Rice 1968



path-independent J

HRR fields



J -controlled tip fields

Testing



elastic-plastic fracture

Final Takeaway



Rice's idea turns crack-tip danger into a measurable energy quantity.